MIL-P-46420B (PA)
10 July 1975
SUPERSEDING
MIL-P-46420A (MU)
30 April 1964

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M57
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing of one type of percussion primer designated as Primer, percussion, M57.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Method of
MIL-P-223	- Powder, Black
MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-P-20449	- Primer, Percussion, M61, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1168	- Ammunition Lot Numbering
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

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DRAWINGS

ARMY

8838009 - Primer, Percussion, M57 Assembly
7548631 - Carton, Packing, Ammunition, for Primers, Percussion M57 and T77
7548632 - Box, Packing, Ammunition, for Primers, Percussion M57 and T77

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Percussion primer.-The percussion primer shall comply with all requirements specified on Drawing (dwg.) 8838009, and with all requirements specified in applicable specifications.

3.3 Primer assemblies without black powder charge.

3.3.1 Non-functioning.-The primers shall not function under an applied energy value of 47.52 inch ounces minimum (min.), when tested as specified in 4.3.1.1.

3.3.2 Functioning.-The primers shall function under an applied energy value of 159.20 inch ounces maximum (max.), when tested as specified in 4.3.1.2.

3.4 Primer assembly with black powder charge.

3.4.1 Functioning.-If the percussion element fires, the primer assemblies shall function without audible hangfire, when tested as specified in 4.3.2.

3.4.2 Pressure resistance.-The primer assemblies shall withstand a pressure of 48,000 minus 2,000 pounds per square inch (psi), piezo-equivalent pressure (pep), without evidence of rupture, blowback around the primer element, or any part of

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the primer becoming detached, when tested as specified in 4.3.2.

3.5 Moisture content of black powder.-The moisture content of black powder at the loading station, at the time of loading, shall not exceed 0.5 percent when tested as specified in 4.3.3.

3.6 Workmanship.-All parts shall be free of burrs, chips, dirt, grease, rust, corrosion and other foreign materials. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.2 Inspection Provisions

4.2.1 Inspection Lot Formation.-The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure

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which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Component parts bearing the same lot interfix number from one manufacturer.
b. Black powder from not more than one lot.
c. Percussion elements from not more than one lot.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects when listed shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Sampling plans shall be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Assembly, prior to assembling percussion element into head (see dwg. 8838009).

Categories	Defects	Method of Inspection
Critical:		
	1. Any flash hole missing from body	Gage
Major:		
	101. Flash hole in head completely obstructed	Visual
Minor: None difined.		

4.2.2.2 Assembly, prior to assembling battery cup and firing plug (see dwg. 8838009).

Categories	Defects	Method of Inspection
Critical: None defined.		
Major:		
	101. Percussion element missing, cocked or inverted	Visual

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Minor:
201. Seal between percussion element and head missing or incomplete....Visual
202. Evidence of poor workmanship (see 3.6).....Visual

4.2.2.3 Assembly, prior to assembling of liner (see dwg. 8838009).

Categories	Defects	Method of Inspection
Critical:		
1.	Firing plug or battery cup above flush with surface of head.....	Straight Edge
Major:		
101.	Firing plug missing or inverted...	Visual
102.	Battery cup loose.....	Manual
Minor:		
201.	Evidence of poor workmanship (see 3.6).....	Visual

4.2.2.4 Assembly, prior to assembling of diaphragm (see dwg. 8838009).

Categories	Defects	Method of Inspection
Critical: None defined.		
Major:		
101.	Weight of charge.....	Balance
*102.	Any flash hole not completely sealed.....	Visual
103.	Liner fails to cover all holes....	Visual
Minor:		
201.	Foreign matter in charge.....	Visual

*Sketches 1, 2 and 3 of Figure 1 indicate acceptable primers. Furthermore, a primer with holes displaying the conditions 4,

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5, or 6 may be classed acceptable provided that the visible liner is thoroughly impregnated with paint or lacquer and not more than 20 percent of the holes have this condition.

4.2.2.5 Assembly (see dwg. 8838009).

Categories	Defects	Method of Inspection
Critical:	None defines.	
Major:		
101.	Sealant of diaphragm missing or inadequate	Visual
102.	Diaphragm punctured or improperly assembled	Visual
Minor:		
201.	Total length, max	Gage
202.	Diameter of flange, max	Gage
203.	Diaphragm above flush with end of body	Visual
204.	Marking misleading or unidentifiable	Visual
205.	Evidence of poor workmanship (see 3.6)	Visual

4.2.2.6 Carton, Prior to sealing (see dwg. 7549631).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	None defined.	
Minor:		
201.	Any packing material missing	Visual
202.	Separator damaged to extent that primers touch each other	Visual
203.	Primer missing	Visual

4.2.2.7 Carton, sealed (see dwg. 7548631).

Categories	Defects	Method of Inspection
Critical:	None defined.	

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Major: None defined.

Minor:

201.	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal box completely.....	Visual
202.	Carton torn, cut or punctured.....	Visual
203.	Marking missing, misleading or unidentifiable.....	Visual
204.	Head end of carton not identifiable.....	Visual
205.	Contents loose.....	Manual

4.2.2.8 Sealed barrier bag (see dwg. 7548631).

Categories	Defects	Method of Inspection

Critical: None defined.

Major:

101.	Seal improper or incomplete.....	Visual
102.	Bag punctured, torn or cut.....	Visual

Minor: None defined.

4.2.2.9 Packing box, prior to sealing (see dwg. 7548632).

Categories	Defects	Method of Inspection

Critical: None defined.

Major:

101.	Carton missing.....	Visual

Minor: None defined.

4.2.2.10 Packing box, sealed (see dwg. 7548632).

Categories	Defects	Method of Inspection

Critical: None defined.

Major:

101.	Box broken, board loose, or protruding nail.....	Visual

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102. Strapping missing, broken or loose.....Visual-Manual

Minor:

201. Marking misleading or unidentifiable.....Visual

202. Strapping improperly engaged or mislocated.....Visual

203. Contents loose.....Manual

4.2.3 Testing

4.2.3.1 Primer assembly without black powder charge.

4.2.3.1.1 Non-functioning (see 3.3.1) - Critical defect

4.2.3.1.1.1 Beginning with the first lot produced, and continuing until three consecutive lots have complied with the acceptance criteria specified, a sample of 150 assemblies shall be randomly selected from each lot for the test. If any sample fails to comply with requirement the lot shall be rejected. The test shall be performed as specified in 4.3.1.1.

4.2.3.1.1.2 After three consecutive lots have met the criteria of 4.2.3.1.1.1, sample of 50 assemblies shall be randomly selected from each lot for this test. If one or more primers fails to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.1.1 using equipment in accordance with 4.2.4. Primers subjected to this test shall be destroyed.

4.2.3.1.2 Functioning (see 3.3.2) - Major defect

4.2.3.1.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified two hundred and fifty samples shall be randomly selected from each lot for this test. If three or more primers fail to comply with the applicable requirements, the lot shall be rejected. If one or two defects are found, a second sample of five hundred primers shall be tested. If the combined number of defectives in both the first and second sample is three or more, the lot shall be rejected. The test shall be performed as specified in 4.3.1.2 using equipment in accordance with 4.2.4.

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4.2.3.1.2.2 After three consecutive lots have met the criteria of 4.2.3.1.2.1, a sample of 125 primers shall be randomly selected from each lot for this test. If two or more primers fail to comply with the applicable requirement, the lot shall be rejected. If only one defect is found, a second sample of 250 primers shall be tested. If the combined number of defectives in both the first and second sample is two or more, the lot shall be rejected. The test shall be performed as specified in 4.3.1.2 using equipment in accordance with 4.2.4.

4.2.3.2 Primer assembly functioning and pressure resistance (see 3.4) - Major defect - A sample of 25 primers shall be selected from each lot for this test. If any primer fails to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.2.

4.2.3.3 Check test for deterioration of Primer, Percussion (M57 Assembly).-If the total elapsed time between original acceptance of the M61 primer lot and the assembly of that lot into the M57 primer exceeds two years, or if the M61 primers have been subjected to adverse conditions for any period of time since previous test, the M61 primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-20449 immediately before the M61 primer lot is assembled into the M57 primers (see 6.3).

4.2.3.4 Moisture content of black powder (see 3.5) - Major defect - A sample of one primer body, prior to assembly to the head, shall be randomly selected from each eight hours production. The black powder shall be tested as specified in 4.3.3 and any failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample.

4.2.3.5 Heat seal test of barrier bag (see dwg. 7548631) Major defect, - Sampling, acceptance criteria and test procedures for this test shall be as prescribed in Specification MIL-P-116 for the "Heat Seal Test".

4.2.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2 and 4.2.3 (see 6.2).

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4.3 Test methods and procedures

4.3.1 Assembly, prior to loading of black powder charge.

4.3.1.1 Non-functioning.-The primer assembly prior to the loading of the black powder charge, shall be assembled into the test fixture (see 4.2.4). A steel ball weighing a min. of 7.92 ounces shall be dropped from a height to give the required impact force (approximately 6 inches).

4.3.1.2 Functioning.-The test shall be performed as specified in 4.3.1.1 except that the weight of the ball shall be 7.96 ounces max., and the height shall be adjusted to give the required impact force (approximately 20 inches).

4.3.2 Determination of moisture content of black-powder.- The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223.

4.3.3 Proving Ground.-This test shall be conducted at a Government proving ground in accordance with the applicable Ordnance Proof Manual and Test Procedure. The primers shall be assembled into complete rounds for which they are standard and fired at a pressure of 48,000 minus 2,000 psi (pep).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-The primers shall be preserved and packaged in accordance with dwg. 7548631.

5.2 Packing

5.2.1 Level C.-The cartons shall be packed in accordance with dwg. 7548632.

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5.3 Marking.-Marking shall be in accordance with dwgs. 7548631 and 7548632.

6. *NOTES*

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Card shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall

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include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204 "Inspection Equipment Design" for guidance. The item detail specification number, paragraph number, and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Cost of check test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test. The test shall be conducted at Government expense without cost to the contractor who loaded the M61 primer or the contractor loading and assembling the M57 primer, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., while the primers were under the jurisdiction of either contractor.

6.4 Check test for possible deterioration of percussion artillery primers.-The following tests are designed to prevent the loading of artillery primers which have deteriorated into a major ammunition item. The test shall be performed on

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artillery primers from each lot which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the artillery primers are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in the contract.

6.4.1 Head assembly Functioning (see 3.3.2).—The sampling plan for the test shall be in accordance with Table II. Any primer assembly which fails to comply with the applicable requirement shall be rejected. The test shall be performed as specified in 4.3.1.2, except that the black powder charge shall be separated in a manner that will not damage the assembly to the extent that the validity of the test results would be questionable.

TABLE II

Sample	Sample Size	Combined Samples Size	Acceptance Number	Rejection Number
First	75	75	*	2
Second	75	150	0	3
Third	75	225	0	3
Fourth	75	300	1	4
Fifth	75	375	2	5
Sixth	75	450	2	5
Seventh	75	525	4	5

*No acceptance on first sample.

6.4.2 Moisture content of black powder.—A sample of 25 primers shall be selected for this test. The black powder shall be removed from the body by any suitable means.** The moisture content of the powder shall be determined in accordance with the methods specified in Specification MIL-P-223. If the moisture content of the powder tested exceeds 0.7 percent, the lot of primers shall be rejected.

**Since black powder is hygroscopic, it is advisable to place the powder in a dry, air-tight container immediately upon removal, and to run the determination as soon as possible thereafter.

Custodian:
Army-PA

Preparing activity:
Army-PA

Project Number: 1390-A094

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ARTILLERY PRIMERS
STANDARDS FOR PAINT CONDITIONS IN FLASH HOLES

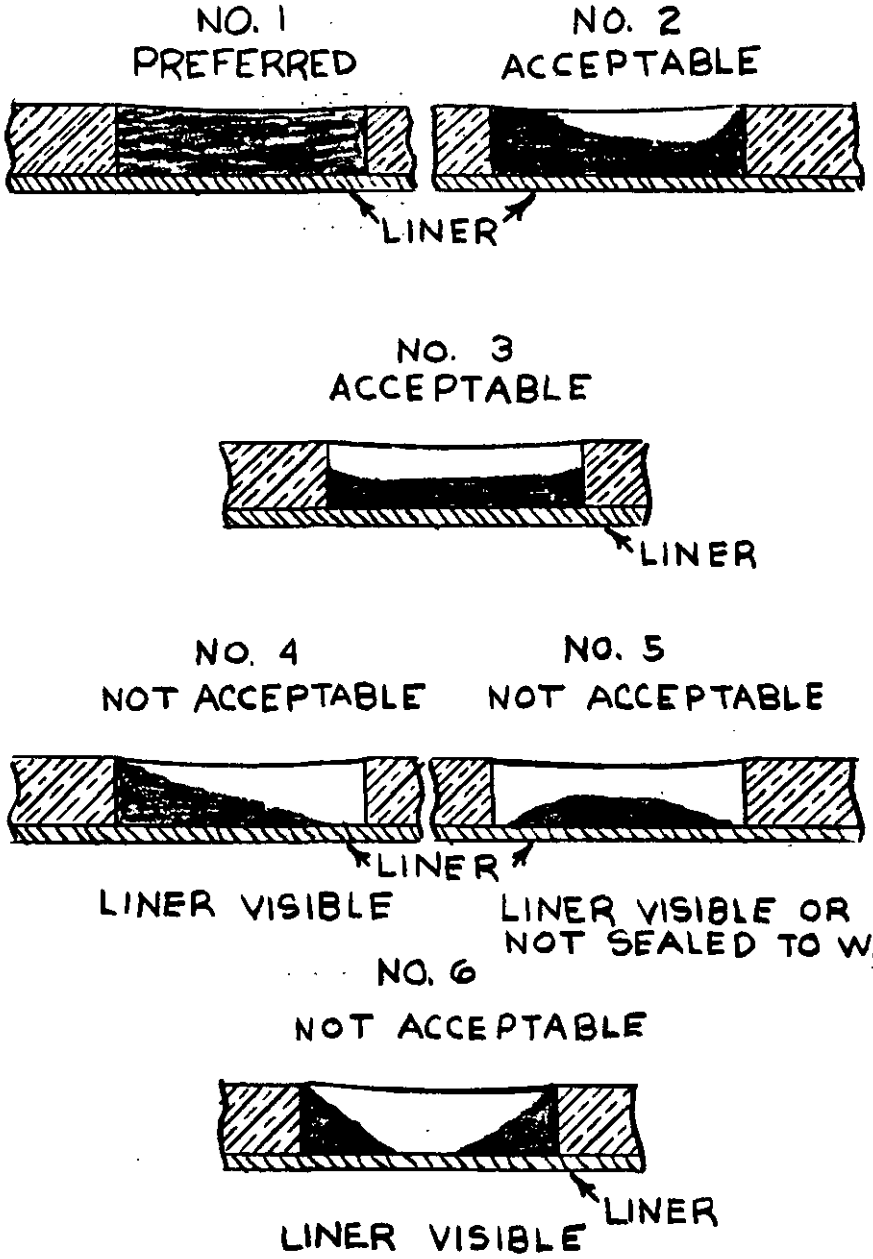


FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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